

1.2 Gaileo's Relativity Principle

To consider mechanical phenomena we need to choose a frame of reference. The laws of motion are in general different across frames.

Finding a frame crucial for simplifying a problem.

For an arbitrary frame, space is inhomogeneous and anisotropic, meaning that where you are located in space matters (positions are not strictly equivalent), and direction also matters. Time is also inhomogeneous in the sense that two instants are not equivalent.

This being said, we can choose frames in an instance where space is homogeneous and isotropic and time is homogeneous; These are called inertial frames.

Immediate references to Lagrangian in an inertial frame. The homogeneity of space and time in an inertial frame implies that the Lagrangian cannot explicitly contain either the radius vector or the time (so L must be a function of v only); Excluding the coordinates are how we account for the homogeneous properties.

But since space is isotropic we say that the lagrangian is a function more precisely of only is magnitude; $L = L(v^2)$

Since the Lagrangian is independent of r , we have $\partial L / \partial r = 0$, and so lagrange's equation is:

$$d/dt(\partial L / \partial v) = 0$$

when $\partial L / \partial v$ is constant. With this we also say v is constant. We conclude that free motion thus has velocity constant in direction and in magnitude (the law of inertia).

For a second inertial frame moving at a constant velocity v relative to the original still inertial frame, the laws of free motion will be the same between them.

Something stronger can be said: These frames are mechanically equivalent. The generalization of this to all inertial frames is Gailileo's relativity principle.

For two reference frames K and K' we have coordinates r and r' , when one moves relative to the other with velocity V , we say that;

$$r = r' + Vt$$

With assumption that $t = t'$